

A Tutorial on Representational Similarity Analysis for Research in Social Cognition

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Representational similarity analysis (RSA) is a simple and widely used technique for comparing relationships between diverse types of measures using a common set of elements. For instance, RSA allows researchers to compare brain and behavioral measures of the same set of participants, or response patterns to the same set of stimuli across different conditions. This technique effectively quantifies relationships between complex, high-dimensional measures even when their measurement units are not directly comparable, contributing to its recent popularity. However, existing tutorials focus on neuroscience use cases, limiting their applicability to social cognitive research. In this tutorial, we cover an accessible introduction to RSA, discussing its strengths and limitations compared to other multivariate methods. We provide commented code and functions in R and Python in the context of two examples that showcase different applications of RSA, illustrating its potential to address questions of broad interest to the field of social cognition.

Keywords: representational similarity analysis, tutorial, social cognition

Social cognitive research aims to understand how people mentally construct and interpret social reality. However, a fundamental challenge is that mental phenomena are complex and not directly accessible (Bless & Fiedler, 2014; Fiske & Macrae, 2012; Greenwald & Lai, 2020). To manage this problem, researchers often use methods that simplify their theories and data (e.g., using 2×2 designs to study

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factors linked to differences in group-level means, summarizing social inferences into a few latent dimensions), making it easier to analyze observable measures that reflect psychological constructs or processes. While this strategy has been highly generative, it risks masking important patterns that are critical to advancing theories of social cognition.

Recent methods now allow researchers to step outside this paradigm. Instead of trying to recover cognitive mechanisms one simplified link at a time, researchers can model the relationships across many variables simultaneously. For example, one can examine how rich, multivariate knowledge structures across trait concepts (e.g., warm, intelligent, funny) and mental states (e.g., affection, contemplation, playfulness) influence each other when forming social inferences (Lin & Thornton, 2023), or how the relations between these inferences differ across group identities (Xie et al., 2021). One popular method that makes this possible is *representational similarity analysis* (RSA), which explicitly models and compares complex, multidimensional data. It first computes similarities within each “space” of measures, and then makes inferences about the relations among different “spaces” by computing the similarities between their patterns, also known as a second-order isomorphism¹ (Shepard & Chipman, 1970). Analyzing similarities in high-dimensional spaces makes for a versatile tool for comparing diverse data types at scale. Crucially, it can compare vastly different data sets generated from different individual minds, collectives, measures, and models.

Due to its flexibility, RSA is now widely used in neuroscience to compare large data sets across different modalities (e.g., fMRI voxels, EEG recordings, behavioral measures within the same set of participants; Kriegeskorte et al., 2008; Nili et al., 2014; Popal et al., 2020). While RSA has also been featured in research on social impressions (Brooks & Freeman, 2018; Cao et al., 2023; Hong & Ratner, 2021; Lin et al., 2021; Stolier et al., 2020; Xie et al., 2021), mental states (Lin & Thornton, 2023; Tamir et al., 2016), and attitudes (de Bruin et al., 2023; Marti et al., 2023; Oktar et al., 2023), all existing tutorials focus on applications in neuroscience and are less accessible to other fields. To address this gap, our tutorial aims to help researchers use and evaluate other work that uses this statistical technique.

In the following sections, we highlight how this technique can advance theories of social cognition. Although mental representations, content, and processes are not directly accessible, examining higher order patterns can uncover novel insights such as shared structural or relational principles. When these structural relationships are not detectable at the level of direct, first-order comparisons, they must be discovered by methods that analyze patterns at the level of second-order similarities. This is often the case when researchers want to investigate mental representations.

1. Consider a hypothetical study about *Blorgorffs* and *Groshmaras*, two alien species from planets B and G. On planet B, *Blorgorffs* survive by being blue, avoiding predators who confuse blue with other hues. If *Groshmaras* shared this trait, then there would be a **first-order isomorphism**. But on planet G, *Groshmaras* survive by being greasy, slipping through the webs, traps, and jaws of their predators. A first-order comparison suggests that these species are very different. Yet their interactions with other species reveal a **second-order isomorphism**: They navigate interspecies interactions in analogous ways. Relations between *Blorgorffs* and species on planet B resemble relations between *Groshmaras* and species on planet G. Despite their differences, their shared patterns of interactions with other species on their respective planets may correspond to parallel social or ecological roles.

WHY COMPARE PATTERNS?

RSA² is a flexible statistical technique that allows researchers to compare patterns across different types of data. While the term *representation* in RSA stems from its application in studying how systems encode external information, the technique itself is agnostic to theoretical assumptions about the meaning of *representations*, and—in its most general form—focuses on analyzing higher order similarity structures in complex measurement spaces. Rather than examining direct, first-order relationships, RSA reveals how patterns of relationships in one data set correspond to patterns in another. (See Supplementary Materials for an extended discussion around the study of representations.)

This approach has proven particularly valuable in fields like neuroscience, where critical information may not be directly encoded in the firing activity of individual neurons, but in the patterns of synaptic connections and correlated activations between neurons (Kriegeskorte et al., 2008; Kriegeskorte & Diedrichsen, 2016). Similarly, in cognitive science, researchers use RSA to compare how artificial intelligence models and humans process information by examining similarities in their “internal representations” (e.g., how they weigh and integrate different inputs; Sucholutsky et al., 2023).

Yet the statistical approach underlying RSA has broad applications across fields. In its most general form, RSA estimates how patterns of relations in one data set map onto patterns of relations in another. In social cognition, this is particularly valuable because social judgments are inherently relational and context-dependent. When researchers use numerical rating scales to evaluate social judgments (e.g., “On a scale from 1 (*not at all*) to 7 (*very much*), how trustworthy does this person appear?”), the meaning of these measurements can shift dramatically with context. For instance, the “cheerleader effect” shows how evaluations of individuals reliably change depending on whether they appear alone or in a group (Carragher et al., 2021; Walker & Vul, 2014), suggesting that the meaning of measurements used in direct, “first-order” analyses is context-dependent.

This context sensitivity makes RSA particularly useful for social cognition research, given the relational nature of social cognition. For example, to understand how stereotypes about societal groups influence individual perceptions (Kawakami et al., 2017; Macrae & Bodenhausen, 2000), researchers can use RSA to compare collective representations (shared stereotypes, schemas, and scripts associated with group identities) with individual-level representations (such as trait impressions and personal judgments of individual group members; Howard, 1994; Morgan & Schwalbe, 1990). RSA’s ability to quantitatively compare these representations makes it a valuable and versatile tool.

In the following section, we illustrate how RSA can be applied to address this research question, discussing its strengths and limitations relative to existing methodologies.

2. Not to be confused with “response surface analysis,” which is used to probe the extent to which combinations of two predictive variables relate to an outcome variable.

MANAGING COMPLEXITY IN REPRESENTATIONS

How do societal stereotypes influence individual perceptions? Depending on how researchers scope this question, the decision space can become overwhelmingly complex. The classic solution has been to avoid complexity by using narrow research scopes. For instance, instead of sampling 50 stereotypes and 40 trait judgments of 30 societal groups, researchers may rely on assumptions of universality (e.g., how stereotypes influence perceptions for *Blorgorffs* should generalize to others) or centrality (e.g., prioritizing stereotypes about some categories like age, gender, and race) to focus on a few stereotypes, traits, and groups. These targets are usually derived using multivariate data reduction techniques (e.g., latent variable modeling, eigenvalue decomposition analysis, cluster analysis) to reduce the set of groups, stereotypes, and measures to a small handful that are relevant, representative, or statistically justified (Bollen & Bauldry, 2011; Borsboom et al., 2021; Kruglanski & Higgins, 2016). This approach has generated highly interpretable models, such as the two-factor model of intelligence (Spearman, 1904), the Warmth–Competence model of stereotype content (Fiske et al., 2002, 2007), the Valence–Dominance model of face perception (Oosterhof & Todorov, 2008), and the Opportunity–Threat model of social relevance (Lassetter et al., 2021).

Although these strategies are efficient, useful, and theoretically generative, they raise additional questions about the validity of the theoretical assumptions (e.g., Rohrer & Arslan, 2021; Schimmack, 2021), the appropriateness of the data reduction solution (e.g., McNeish & Wolf, 2020; Revelle, 2024), and the replicability and generalizability of the models (Nosek et al., 2022; Yarkoni, 2022). As a case study, two-dimensional models that summarize variance in trait impressions are highly influential in social cognition research, but recent work indicates that they may not generalize to different contexts (Abele et al., 2021; Koch et al., 2016), analyses (Lin et al., 2021), cultures (Jones et al., 2021), groups (Hutchings et al., 2024; Xie et al., 2021), stimulus sets (Sutherland et al., 2013, 2016), or less constrained research designs (Nicolas et al., 2022).

To some extent, these concerns connect to broader methodological issues in psychology, which we highlight in the next section. While these discussions are not the focus of this article, they provide context for the utility of RSA. Specifically, RSA serves as a flexible computational technique that can complement existing methods in social cognition research.

REPRESENTATIONAL SIMILARITY ANALYSIS

One common strategy exploited by most data reduction methods—and that is a precursor to these methods—involves constructing a similarity matrix (usually a correlation, covariance, or distance matrix) to capture the similarity or dissimilarity between pairs of features (Kriegeskorte et al., 2008; Nili et al., 2014). For example, researchers studying facial impressions may collect judgments of faces on various traits (e.g., trustworthy, intelligent, aggressive, attractive), and find that some trait judgments are strongly correlated, such that if any person’s face is

rated as very friendly, it will likely be rated as very trustworthy (Lin et al., 2021; Oosterhof & Todorov, 2008; Sutherland et al., 2013). A similarity matrix captures these similarities across all trait-pairs in the “trait space” of facial first impressions (Stolier et al., 2018; Xie et al., 2021).

Similarity matrices are often high-dimensional. Data reduction methods (e.g., principal component analysis) simplify these matrices by summarizing correlated measures into a smaller set of dimensions. Although this approach is useful for extracting broad patterns from complex data, its application assumes that data can meaningfully be collapsed into fewer dimensions without losing essential information. This assumption is not always justified in social cognition research (Lu & Lin, 2024). The advantage of RSA is that it can compare similarity matrices with or without data reduction, allowing researchers to preserve complexity in the data and avoid assumptions associated with data reduction.

While data reduction techniques and RSA are often used independently in social cognition research to different ends, they are not mutually exclusive and can complement each other. For example, factor analysis can guide hypothesis generation by identifying key factors or reducing the noise in complex data, whereas RSA allows researchers to compare high-dimensional data sets more comprehensively. They can be used directly in combination, for instance performing RSA by calculating pairwise similarities of principal components. However, in cases where data reduction risks oversimplifying the data, RSA offers a way to preserve and analyze the original, high-dimensional relationships.

RSA ALLOWS COMPARISON OF COMPLEX, HETEROGENEOUS DATA

The main advantage is that RSA can explicitly model and compare high-dimensional data across different measurement units³ or levels of analysis. For example, researchers may investigate whether stereotypes about societal groups influence spontaneous impressions of strangers. One approach to this question relies on a “first-order” comparison—if women are stereotyped as warmer than men, researchers might test whether the distribution of “warmth” ratings is statistically higher on average for women than for men. A different approach to this question probes patterns of relations—if “women who are perceived as attractive are also more likely to be perceived as warm, but perceived attractiveness and warmth are unrelated for men,” then this “second-order isomorphism” reveals a relational difference across perceived traits that is associated with stereotypes about women and men. Taking this further, such a second-order relational analysis can extend well beyond warmth and attractiveness. If researchers wanted to make general claims about stereotype representations and trait impressions, then rather than focusing on any pair of traits, they may need to justify their broader claim with analytic techniques that can handle comparisons among many traits. Thus, RSA

3. RSA overrides the “raw” measurement units (e.g., BOLD signal, EEG scalp voltage, scale of behavioral measures) because it is based on second-order relationships that carry represented information underlying pairwise patterns of relations. This enables comparison of measurements across different modalities, such as when measurements are taken from different scales.

allows researchers to more comprehensively compare stereotypical and individual representations of many items (e.g., traits), rather than focusing on just one example pair of traits as traditional methods might do.

To clarify this distinction, consider two example research designs (Figure 1). During study design (Figure 1A), researchers may collect trait ratings from many participants on stimuli such as photographs of faces categorized into different societal groups, sampling from distributions of trait judgments in social impressions (e.g., trustworthy, warm, attractive, competent, strong), societal groups (e.g., Black, East Asian, Latine, White men and women), and individual stimuli for each group (e.g., photos categorized into each race-by-gender group). Researchers must decide how broadly to sample and how to handle their data.

During the data engineering phase, researchers decide how to structure their data for analysis. A classic approach that aims to investigate first-order relationships will typically use data reduction methods (e.g., factor analysis, principal component analysis) to reduce the dimensionality of the data (Figure 1B, top) and link each dimension to outcome variables. The goal is to simplify the design of experimental manipulations that test causal processes, maximize differences across conditions (e.g., by choosing orthogonal or distinct trait dimensions or societal groups), and support significance tests of aggregated between-condition effects (Figure 1C, top). For example, researchers may obtain stereotypical ratings of women on various traits (e.g., warm, friendly, attractive), summarize these ratings with a single dimension of “warmth,” and then use scores on the warmth dimension as a predictor in univariate linear models to estimate its relationship with an outcome (e.g., perceivers’ accuracy of categorizing these women into different age groups).

In contrast, RSA can model multidimensional structures of representations with or without data reduction. Researchers choose what comparisons are relevant. To illustrate RSA’s flexibility, in Figure 2 we highlight three examples to showcase RSA. The first approach uses traits as the grouping factor of items whose similarities are computed (see Figure 1B, bottom, and Figure 2A), making the similarity across each trait-pair (e.g., warm~competent) the unit of analysis common to different matrices. In this case, two similarity matrices—one from trait ratings of individual faces and another from trait ratings of societal group labels—can be compared (Figure 1C, bottom) to estimate the representational similarity between “facial-impression space” and “stereotype-knowledge space” (Xie et al., 2021).

A second approach is to use stimuli as the grouping factor, so that the similarity of each stimulus-pair (e.g., LW1.png~BM1.png) is the unit of analysis (Figure 2B). The logic is that if two stimuli are evaluated similarly in their trait impressions, then these similarities may arise from similarities in the stimuli at other levels of representation, such as visual features (e.g., hue, luminance, facial morphology; Keles et al., 2021). To test this, researchers construct another similarity matrix using features of that space (e.g., hue) to compare the representational similarity between “visual-feature space” and “facial-impression space.”

A third approach is to use participants as the grouping factor, so that the similarities of pairs of participants (e.g., Jimmy and Molly) become the unit of analysis

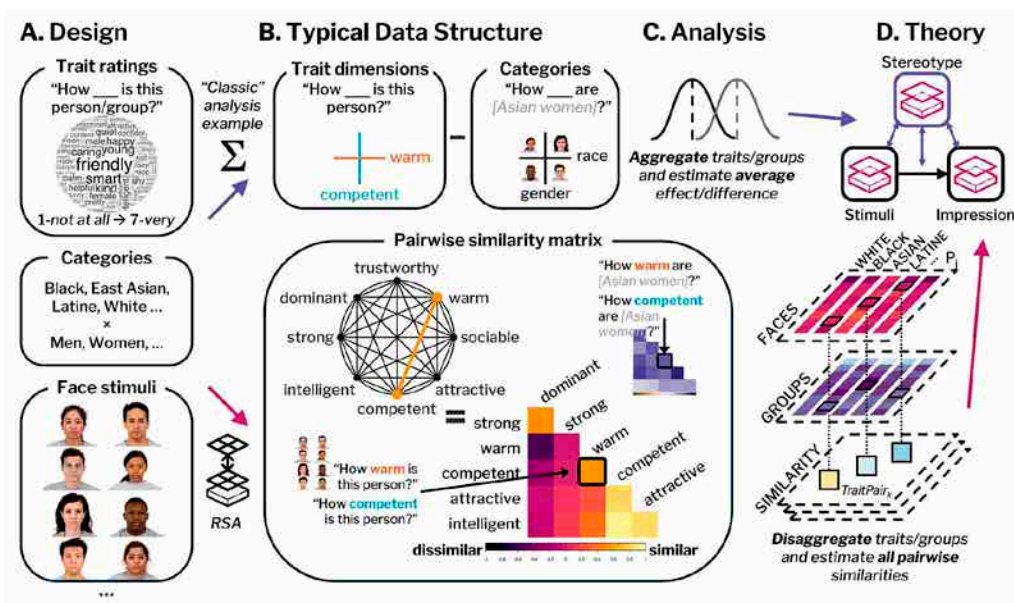


FIGURE 1. Example of two approaches to studying the question, "Do societal stereotypes influence trait impressions?" See main text for detailed description of each panel.

(Figure 2C). This approach is sometimes referred to as "intersubject representational similarity analysis" (G. Chen et al., 2017; P.-H. A. Chen et al., 2020) and can be used to capture individual differences in representations. For instance, do two individuals who evaluate Asian women in a similar way also complete another measure similarly (e.g., endorse stereotypes to the same extent)?

In all cases, at the final step of RSA (Figure 2D), the similarity of similarity matrices is inferentially estimated using parametric or nonparametric tests (L. Chen, 2023; Dutilleul et al., 2000; Viviani, 2021).

Both classic and RSA approaches contribute to theories of social cognition (Figure 1D). The classic paradigm simplifies experimental design, facilitating first-order analyses that isolate constructs to test causal mechanisms. Yet oversimplification is a concern when, characteristic of real-world social cognition, multiple overlapping factors interact in complex ways. RSA complements classic approaches by directly modeling high-dimensional representations, helping researchers uncover systematic patterns and obtain more robust and generalizable effect estimates. For example:

RSA Can Enable Stronger Tests of Group Differences. Researchers have used RSA to compare the representational spaces of normative gender ideals and self-perceptions across different groups of people. Trait representations associated with normative ideals of binary gender categories align with self-representations

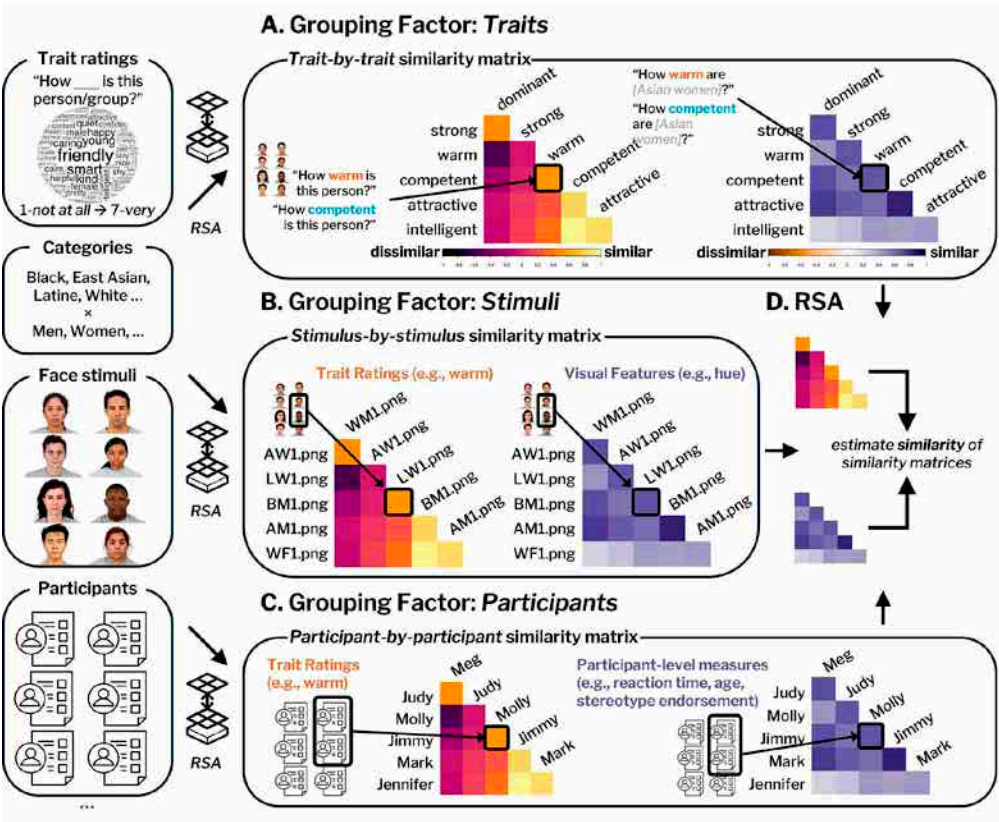


FIGURE 2. Examples of multiple ways to conduct representational similarity analysis, with (A) traits as the grouping factor, (B) stimuli as the grouping factor, or (C) participants as the grouping factor of items common to different matrices. In each case, RSA estimates the similarity of similarity matrices (D).

in gender-conforming women and men—but not in gender-diverse people (Xie et al., 2025). Critically, this divergence only emerges at the level of structural (i.e., second-order) comparisons and is overlooked by classic analyses of mean trait ratings, showing how RSA can reveal structural patterns.

RSA Can Be Used to Test Competing Hypotheses. Consider a study that tests competing hypotheses about stereotype formation. Stereotypes about immigrants may be shaped by cultural norms (H_1), intergroup contact (H_2), or neither (H_0). Researchers collect data from different regions (e.g., neighborhoods, census-based metropolitan areas) on stereotypical impressions of immigrants (e.g., trustworthiness, work ethic, education), cultural norms and attitudes toward immigration, and intergroup contact measures. Using each pair of regions as units of analysis, researchers can examine whether regional similarities in immigrant stereotypes

align more with regional similarities in cultural norms (supporting H_1), more with regional similarities in intergroup contact patterns (supporting H_2), or neither (failing to reject H_0).

RSA Can Test Causal Relationships. RSA can be leveraged to test causal relationships in representational spaces, particularly when paired with experimental designs that enable double dissociation. For example, in a study on belief structures, participants are randomly assigned to two experimental conditions—to watch a charismatic podcaster discuss topics, or to interact with strangers in online video calls to discuss the same topics. Afterward, participants report their endorsement of each belief. By constructing a similarity matrix from pairs of beliefs—representing the extent to which any pair of beliefs are similarly endorsed—researchers can compare how the experimental conditions shape belief representation. In the podcaster condition, belief endorsements about topics discussed by the podcaster may cluster more tightly, reflecting greater similarity in views. In the interaction condition, endorsements about topics debated in the video call may be more dispersed, reflecting greater diversity in views. This approach leverages RSA to test causal relationships in belief representations. A double dissociation could emerge if the podcaster condition increases clustering of discussed beliefs while the video call condition increases differentiation. By combining RSA with experimental design, researchers can isolate condition-specific effects on belief representations.

RSA DOES NOT ASSUME CLEAN FACTOR STRUCTURES

RSA also serves as a measurement comparison approach in complex measurement spaces where clean factor structures are not always present. For example, the Social Dominance Orientation scale has three subfactors (Ho et al., 2015). Items in this scale uniquely and cleanly load on each subfactor. Cross-loading, where items load strongly on multiple factors, is reduced by the scale development process. When researchers use this scale in a new sample or context, they confirm whether the scale works similarly (as is considered best practice; Kane, 2013) with confirmatory factor analysis, testing whether the items load cleanly on these subfactors as expected.

However, clean loadings are not always the case. As discussed above, people form spontaneous impressions on many traits, such as attractiveness or intelligence, and ratings on these traits do not load cleanly on any single dimension. Attractiveness, for instance, loads on both Valence and Dominance to a meaningful extent, as do other common traits (Jones et al., 2021; Lin et al., 2021; Sutherland et al., 2013). This means that confirmatory factor models of independent factors of Valence and Dominance (or other factors) will fit far below acceptable thresholds (Hsu et al., 2015). Unlike in scale development, cross-loading items cannot simply be eliminated to maximize model fit: People *do* rate others on attractiveness, naturally and spontaneously (Zebrowitz, 2006). This issue is prevalent in

complex measurement spaces such as social trait inferences about people (Hester & Hehman, 2023; Koch et al., 2016). To remove “poorly fitting” items from measurement is to conform nature to measurement, rather than measurement to nature (Revelle, 2024), which must be pursued if we wish to represent the real, messy world in our models.

RSA provides an alternative measurement comparison approach in such a scenario because it does not assume any factor structure. It has been used to compare whether different race and gender targets are perceived in the same way (Xie et al., 2021), or whether individual versus regional prejudices are distinct (Suliteanu et al., 2024). Given complex measurement spaces, RSA balances the trade-off between “interpretable” theories and methodologies (aiming for parsimonious explanations of mental processes) and data-driven methods (aiming for accuracy at the cost of explanatory power; Yarkoni & Westfall, 2017). When the assumptions of classic data reduction solutions are violated, it may indicate that the data truly are high-dimensional and complex, and reducing them to lower dimensional spaces results in a loss of important information. Yet at the other extreme, “black box” computational methods that capture complex nonlinear effects reduce interpretability (Keles et al., 2021; Yarkoni & Westfall, 2017). RSA complements existing research methods by interpolating between interpretability and complexity: Researchers choose which representations to compare and select the items whose representations are compared, allowing RSA to connect and contribute to theory.

ASSUMPTIONS AND LIMITATIONS OF RSA

Despite its advantages, RSA makes several assumptions of its own about the relationship between representational spaces. When these assumptions are violated, RSA can fail to detect true relationships, increasing Type II error, or overestimate true relationships, increasing Type I error. This can bias model comparisons if an incorrect model satisfies the assumptions better than the correct one. These challenges have been documented in comparing neuroimaging data to other modalities (Cai et al., 2017; L. Chen, 2023; Diedrichsen & Kriegeskorte, 2017; Dujmović et al., 2022) and in evaluating computational models (Dujmović et al., 2022; Hermann & Lampinen, 2020). We discuss how these limitations affect RSA in social cognitive research and provide recommendations for addressing them.

RSA ASSUMES ALL PAIRWISE RELATIONSHIPS MATTER EQUALLY: WHAT IF ONLY A SUBSET IS IMPORTANT?

Consider a scenario where stereotypes only affect impressions of faces for the relationship between a few trait-pairs, such as trustworthy and dominant. RSA may fail to find a significant relationship between “stereotype-knowledge space” and “facial-impression space” because this specific relationship (trustworthy~dominant) is diluted by many other nonsignificant relationships included in the representational

space (e.g., attractive~competent, intelligent~dominant, trustworthy~friendly, sociable~warm).

To address this, researchers can use theory-driven approaches to identify a representative trait-space during study design (L. Chen, 2023) and apply methods that allow flexible weights for different items in RSA (e.g., partial least squares regression; Viviani, 2021). For example, researchers can first identify spontaneous trait descriptions through open-ended tasks (Asch, 1946; Klapper et al., 2016; Park, 1986; Sutherland et al., 2018) and use these traits in the trait-space.

If certain trait-pair relationships are expected to be more important, this can be modeled in the final step of RSA (e.g., using mixed-effects models to account for unequal contributions of items). We discuss this and other approaches in the Supplementary Materials, and we illustrate one approach in the tutorial section.

RSA ASSUMES THE CANDIDATE REPRESENTATIONAL SPACE IS RELEVANT AND REPRESENTATIVE

Another issue is the selection of items for the matrices in RSA, which can introduce bias. For instance, if researchers want to test how gender stereotypes shape trait impressions, they might select traits that are highly gender-stereotypical to construct their spaces. This can lead to an overestimated relationship between these spaces because RSA is biased by how the items were selected—oversampling traits that maximize gender differences and undersampling other relevant traits.

However, this limitation is not unique to RSA. Maximizing variance on some dimension to observe a significant effect is common in traditional research designs. RSA can address this issue more effectively than classic methods if used in a principled way, such as selecting representative and relevant spaces informed by descriptive research (e.g., collecting unconstrained trait descriptions in early stages of research design) to avoid distorting the representational space to confirm a specific hypothesis.

RSA IS SUSCEPTIBLE TO THE ACCUMULATION OF MEASUREMENT ERROR

RSA can be vulnerable to noisy measurements. When data contain noise, the (dis)similarity metrics used to construct matrices can accumulate measurement error, leading to biased estimates of the similarity between these matrices (Cai et al., 2019; Walther et al., 2016). To mitigate this problem, researchers should choose metrics appropriate to the relationships they want to test,⁴ minimize sources of

4. Selecting an appropriate metric often requires formal evaluation of the research context and properties of the data. For example, Euclidean distance is appropriate for assessing raw differences based on magnitude and works well when variables across data sets are measured on the same scale and have similar ranges (e.g., Likert-scale ratings across items). Correlations are appropriate for assessing the relative pattern of responses regardless of absolute values (i.e., magnitude). Mahalanobis distance can account for the covariance structure of the data, and is useful when measures have different units, scales, or are highly correlated. Rank-based metrics such as Spearman's rank

correlated errors during study design, and consider using cross-validation and regularization to reduce bias, even if it means increasing variance (Diedrichsen et al., 2021; Nili et al., 2014).

For example, during study design, researchers can randomize the presentation of traits and stimuli to minimize serial correlations from items presented in proximity. Counterbalancing the order of conditions and stimuli across participants can further reduce correlated measurement error. Common similarity/distance metrics (e.g., Pearson, Spearman, and Kendall correlations, Euclidean distance, absolute distance, cosine similarity, Mahalanobis distance) tend to be biased, especially when correlated measurement error is likely (e.g., when conditions cannot be counterbalanced, or stimuli cannot be randomly presented; Popal et al., 2020). Researchers may want to apply transformations to their (dis)similarity metrics to help reduce the influence of noise, correlated errors, and unknown confounds (Diedrichsen et al., 2021; Dujmović et al., 2022; Meng et al., 1992). For instance, researchers can apply Fisher- z transformations to correlation metrics, subtract known sources of noise and common method variance from their metrics by combining RSA with regression (i.e., removing the common factors and comparing residuals), and directly account for other related variables in the model. Finally, theory-driven RSA can mitigate many of these concerns. In the Supplementary Materials, we provide additional information and resources about more technical methods that can address these limitations, and the following tutorial demonstrates the use of some of these strategies.

(DIS)SIMILARITIES CAN BE ASYMMETRIC

RSA is typically implemented in a way that assumes symmetry in pairwise (dis)similarities (A's similarity to B equals B's similarity to A). However, researchers may want to relax this assumption by using directional distances rather than absolute (dis)similarities. For example, past work shows that the transition probability between different mental states can be asymmetric (e.g., people expect the transition from "drunk" to "sleepy" to be more likely than the reverse; Thornton & Tamir, 2021; Thornton et al., 2023), and classic work shows that Luxembourg seems more psychologically similar to France than France to Luxembourg (Kahneman & Tversky, 1973). When asymmetries are expected or theoretically important, researchers should consider directional approaches to RSA that require careful handling of duplicated degrees of freedom. We discuss these technical considerations and their solutions in detail in the Supplementary Materials.

Having introduced RSA, we now turn to a step-by-step tutorial for conducting RSA with researchers' own data. We showcase two applied examples that use RSA to test different questions about how social inferences are drawn from facial appearance.

correlation are suitable for ordinal data or when the hypothesis involves nonlinear relationships. Beyond RSA, exploratory methods such as multiscale graph correlation can be useful for exploring and quantifying patterns across disparate data modalities without relying on predefined similarity metrics; it can discover complex nonlinear relationships at multiple scales by using graph structures to compare data sets (Vogelstein et al., 2019).

TUTORIAL

MOTIVATING EXAMPLE

People spontaneously form rich inferences about personality traits from looking at someone. What drives these inferences? Researchers have studied the contribution of various factors to social perception, such as how target characteristics (e.g., visual appearance; Jaeger & Jones, 2022; Keles et al., 2021) and perceiver characteristics (e.g., identity, learned stereotypes; Oh et al., 2019; Xie et al., 2019) interact to influence impression formation (Hehman et al., 2018). Given that multiple streams of information are available in real life, it is crucial to consider how different sources of information shape first impressions.

Traditional methods like ordinary least squares regression or structural equation modeling can face issues such as multicollinearity, overfitting, or unidentifiable models when trying to regress dependent variables on many independent variables. To address these challenges, we use RSA to examine how different sources of information (e.g., societal stereotypes and visual information) contribute to first impressions. Specifically, patterns of similarity may reflect collective representations of social-categorical stereotypes (common mental models, schemas, or knowledge structures shared across participants) mapped to perceptual cues such as visual features (Freeman et al., 2020; Over & Cook, 2018; Stolier et al., 2020).

In the following examples, we first investigate the extent to which people use categorical information (e.g., stereotype knowledge learned in society) to form trait impressions from faces, showcasing the use of RSA to address this question with R code. Here, we examine how stereotype representations (stemming from shared cultural stereotypes) guide social trait inferences of faces across different social categories. Next, we use computational tools and human annotations to explore how perceptual information (e.g., facial structure, facial expressions of emotion, body action, and body shape) contributes to impression formation (Zheng & Lin, 2024), showcasing the use of RSA to address this question with Python code. In this case, we examine how representations based on visual features map onto representations based on social trait inferences. This approach allows us to examine how representations of stereotypes and visual information predict the space of first impressions.

EXAMPLE 1 IN R: HOW MUCH DO STEREOTYPES CONTRIBUTE TO FACIAL IMPRESSIONS?

Step 1: Data Structuring

To compare the “stereotype-knowledge” space with the “facial trait-impression space,” we construct two similarity matrices from two data sets obtained from different participants. The first data set includes trait ratings made on Likert scales (1, *not at all* to 7, *very much*) in response to photos of individual faces across different race $\times$ gender categories (e.g., “How trustworthy is this person?”). The second data set includes trait ratings made on the same scale in response to race $\times$ gender

group labels (e.g., “How trustworthy are [White men]?”). We first load necessary libraries and data sets in R. See [bit.ly/RSAexample1osf] for full script.

```
library(tidyverse)    # data hygiene
library(psych)        # descriptives & fisher-z transform
library(lme4)         # linear mixed-effects models
library(lmerTest)     # fixed effects for lmer models
select <- dplyr::select

# Load datasets
data_faceratings_raw <- read_csv("example1_data_faceratings_raw.csv")
data_stereotypatings_raw <- read_csv("example1_data_stereotypatings_raw.csv")
```

Selecting Similarity Metrics. Next, we choose an appropriate similarity (or distance) metric for each matrix. Common (dis)similarity metrics include Pearson or Spearman correlations, Euclidean distance, Cosine distance, and Mahalanobis distance. Because both data sets are measured on the same interpretable scale (e.g., 1–7 Likert scales) and we care about the relative relationship between trait ratings, we chose the simplest metric—Pearson correlation (with transformations, as described later). Although we use the Pearson correlation here for its simplicity, it is important to note that it is often not the best measure of (dis)similarity—and this decision often requires a more formal analysis (Nili et al., 2014). We extracted values from the lower triangle of the similarity matrix (because the top triangle is mirrored and redundant), excluded the diagonal (which is each trait correlated with itself), and flattened the matrix into “long format.”

In codeblock 1, we construct the similarity matrix for trait ratings obtained from facial impressions (hereafter the *face-trait space*).

In codeblock 2, we repeat these steps to construct the similarity matrix for stereotypical trait ratings obtained from group labels (hereafter the *stereotype-trait space*).

Step 2: Data Transformations

Having constructed similarity matrices for the spaces we want to compare, we will apply data transformations to reduce the influence of noise, correlated errors, and potential confounds.

Fisher-z Transformation. Because we used Pearson correlations, the first step is to apply Fisher-z transformation to the correlation coefficients (Meng et al., 1992; Nili et al., 2014). This normalizes their distribution, stabilizes their variance, and makes them more suitable for parametric statistical tests, because the z scores follow a normal distribution. This was already applied during the construction of the correlation matrices (see codeblock 1).

Subtract the Average Trait-Pair Correlation (Across All Groups) From Each Correlation. One potential confound is that these spaces share a common method (e.g.,

```

# get the names of the items that will be used as the common set of items across both
spaces (i.e., grouping factor)
# in this case, the common set of items are the traits
item_cols <- names(data_faceratings_raw[3:16])

# create empty dataframe to store "face-trait space", the similarity matrix for trait
ratings of faces
facetraitspace <- data.frame()

# write for loop to construct a separate similarity matrix for each race x gender
group
cluster_levels <- levels(as.factor(data_faceratings_raw$group))

for (cluster in cluster_levels){
  temp_data <- data_faceratings_raw %>% # create temporary dataframe
    filter(group == cluster) %>%      # subset data for that race x gender group
    select(all_of(item_cols))          # keep only the columns with the items
  temp_cor <- cor(temp_data, use="complete.obs", method=c("pearson")) # corr matrix
  temp_cor <- as.matrix(psych::fisherz(temp_cor)) # apply fisher-z transformation
  # convert correlation matrix (one diagonal only) to long format, keep var names
  temp_cor <- as.data.frame(as.table(temp_cor))
  long_cor <- subset(temp_cor, match(Var1, names(temp_data)) > match(Var2,
names(temp_data)))
  long_cor[["group"]] <- cluster # assign race x gender identifier
  facetraitspace <- rbind(facetraitspace, long_cor)
}

# rename variables and create traitpair cluster
facetraitspace <- facetraitspace %>%
  mutate(traitpair = paste0(Var1, Var2)) %>%
  rename(corrFaceRatings = Freq)

```

Codeblock 1

eliciting trait ratings using verbal labels). For example, conceptually similar traits like “friendly” and “warm” may have strong relationships in both spaces due to their semantic similarity, and their evaluation on 1–7 Likert scales. To remove this “common method variance,” we subtract the average value of each trait-pair (e.g., friendly~warm) across all race × gender groups from the value of each trait-pair correlation for each group. For example, the subtracted correlation of friendly~warm for Asian men would represent how that relationship is distinctly different for Asian men relative to other groups (Xie et al., 2021). See codeblock 3.

Step 3: Test of Research Hypothesis

We highlight three approaches to test our research hypothesis, starting with the crudest approach (with the highest possibility of violating assumptions and producing biased estimates) and concluding with the most robust method.

```

# create empty dataframe to store "stereotype-trait space", the similarity matrix for
# trait ratings of race x gender groups in the abstract
stereotypetraitspace <- data.frame()

# write for loop to construct a separate similarity matrix for each race x gender
# group
cluster_levels <- levels(as.factor(data_stereotypetratings_raw$group))

for (cluster in cluster_levels){
  temp_data <- data_stereotypetratings_raw %>% # create temporary dataframe
    filter(group == cluster) %>%           # subset data for that race x gender group
    select(all_of(item_cols))              # keep only the columns with the items
  temp_cor <- cor(temp_data, use="complete.obs", method=c("pearson")) # corr matrix
  temp_cor <- as.matrix(psych::fisherz(temp_cor)) # apply fisher-z transformation
  # convert correlation matrix (one diagonal only) to long format, keep var names
  temp_cor <- as.data.frame(as.table(temp_cor))
  long_cor <- subset(temp_cor, match(Var1, names(temp_data)) > match(Var2,
names(temp_data)))
  long_cor[["group"]] <- cluster # assign race x gender identifier
  stereotypetraitspace <- rbind(stereotypetraitspace, long_cor)
}

# rename variables and create traitpair cluster
stereotypetraitspace <- stereotypetraitspace %>%
  mutate(traitpair = paste0(Var1, Var2)) %>%
  rename(corrGroupStereotypeRatings = Freq)

# confirm that the "grouping factor" (or set of items common to both matrices) are
# the same
stereotypetraitspace$traitpair == facetraitspace$traitpair

```

Codeblock 2

We begin by estimating the similarity of similarity matrices using Spearman rank correlation, without removing common method variance. See codeblock 4.

This produces a high correlation of $\rho = .76$, suggesting a strong, positive linear association between the structure of stereotype knowledge and facial trait impressions. However, since no steps were taken to debias this analysis, this similarity estimate may be overinflated.

Next, we subtract a known source of common method variance (e.g., conceptual similarity of traits) from these matrices and conduct Spearman rank correlation again. See codeblock 5.

After subtracting common method variance, the similarity decreases to $\rho = .16$ ($p < .001$), suggesting a slight positive association between these spaces. Yet this approach assumes that all trait-pairs are equally important. In the next approach, we relax this assumption by inputting these matrices into a mixed-effects/multi-level regression model with random intercepts and slopes. See codeblock 6.


```

facetraitspace <- facetraitspace %>%
  group_by(traitpair) %>%
  # for each traitpair, take the average correlation across all groups
  mutate(corrFace_meanTP = mean(corrFaceRatings)) %>%
  ungroup() %>%
  # remove the average from each correlation
  mutate(corrFace_centeredWithinTP = corrFaceRatings - corrFace_meanTP)

stereotypetraitspace <- stereotypetraitspace %>%
  group_by(traitpair) %>%
  mutate(corrGroupStereotype_meanTP = mean(corrGroupStereotypeRatings)) %>%
  ungroup() %>%
  mutate(corrGroupStereotype_centeredWithinTP = corrGroupStereotypeRatings -
    corrGroupStereotype_meanTP)

```

Codeblock 3

```

# combine data from both spaces
data <- full_join(facetraitspace, stereotypetraitspace)

# start with crudest approach - correlation without debiasing
cor.test(data$corrFaceRatings, data$corrGroupStereotypeRatings,
  use="everything", method="spearman")

```

Codeblock 4

```

# estimate spearman correlation after subtracting the average correlation (of each
# traitpair, across all groups) from each correlation
# whatever "remains" is the unique relationship between a pair of traits specific to
# that race x gender group in both spaces
cor.test(
  data$corrFace_centeredWithinTP, data$corrGroupStereotype_centeredWithinTP,
  use="everything", method="spearman")

```

Codeblock 5

```

# the DV, or Level-1 unit of analysis (corrFaceRatings), is the correlation for
# any pair of traits obtained from ratings of faces.
# For example, one observation is the correlation (z-scored) between trustworthy
# and competent for ratings of Asian women's faces.

# The Level-1 predictor is corrGroupStereotypeRatings, the correlation for that same
# pair of traits (e.g., trustworthy~competent, z-scored) obtained from ratings of
# group labels (e.g., the label of Asian women).

model <- lmer(corrFaceRatings ~
  corrGroupStereotypeRatings + corrGroupStereotype_meanTP +
  (1 + corrGroupStereotypeRatings | traitpair), data = data)
summary(model)

```

Codeblock 6

In this model, we again partition out common method variance by including the `corrGroupStereotype_meanTP` term, which is the average correlation for each trait-pair (obtained from stereotypical group ratings) across all groups. This partials out the effect of the conceptual meaning of the trait-pair itself. The coefficient of interest is the fixed effect of the “within-trait-pair” `corrGroupStereotypeRatings` term, which captures the unique relationship of trait-pairs in stereotype-space to face-trait-space as a function of race $\times$ gender membership. This term is included as a random slope nested within trait-pairs, allowing for the possibility that some trait-pairs contribute more to this effect than others.

The results of this final analysis—accounting for common method variance and nonidentical contribution of different items—suggest that the similarity between stereotype-trait space and face-trait space is $\gamma_{.001} = .13$ standard deviations ($p = .001$). Because both matrices are in z score units, the interpretation of the units is straightforward.

EXAMPLE 2 IN PYTHON: HOW MUCH DOES VISUAL INFORMATION CONTRIBUTE TO TRAIT IMPRESSIONS?

In this example, we investigate the extent to which different visual features might be used to form trait impressions, using the trait “warmth” as an example. For the full script, see [bit.ly/RSAexample2Code].

Step 1: Data Structuring

Because visual cues are perceived from individual images, the common items across different matrices are the images themselves. All annotations and ratings were made at the image level. We constructed matrices to represent: (a) facial-structure space, (b) facial-expressions-of-emotion space, (c) body-action space, (d) body-shape space, and finally (e) trait-rating space. We calculated the aggregate rating for each feature of the human-annotated cues, and for trait ratings of images. See the Supplementary Materials for details about data curation.

Step 2: Selecting Similarity Matrices

We considered two commonly used distance metrics: Euclidean distance and cosine distance. Euclidean distance measures the spatial distance between images and is sensitive to the exact values, whereas cosine distance is used when the magnitude of the vectors does not matter, such as with text data represented by word embeddings. We chose Euclidean distance to construct the matrices for facial-structure, facial-expression-of-emotion, body-shape, and trait-rating spaces (because trait ratings may be driven by feature values), but used cosine distance to construct the matrix for body-action space (because these features were annotated using word embeddings from language models, and word embeddings were trained with cosine distances). See codeblock 7.

```

# Import Package
from sklearn.metrics.pairwise import cosine_distances
from sklearn.metrics.pairwise import euclidean_distances

# Function to compute the representational dissimilarity matrices
# using cosine distance for text embedding cues
def cosRDM(data):
    # Calculate the cosine distance matrix
    rdm = cosine_distances(data)
    # Extract the lower triangle of the matrix, excluding the diagonal
    rdm_lower = list(rdm[np.tril_indices(rdm.shape[0],-1)])
    return rdm_lower

# Function to compute the representational dissimilarity matrices
# using euclidean distance for all other cues
def eucRDM(data):
    # Calculate the euclidean distance matrix
    rdm = euclidean_distances(data)
    # Extract the lower triangle of the matrix, excluding the diagonal
    rdm_lower = list(rdm[np.tril_indices(rdm.shape[0],-1)])
    return rdm_lower

```

Codeblock 7

Step 3: Test of Research Hypothesis

To test our hypothesis that the structures of face and body visual features are linked to how people infer the warmth of the targets, we used ordinary least squares regression. We regressed the distances between images (from trait ratings) against the distances between images (based on face or body cues).

We extracted the values in the lower triangle of the trait distance matrix and the feature distance matrices (computed above), converting these matrices into vectors of dependent variables (distances between images based on trait-ratings) and independent variables (distances between images based on facial structures, facial expressions-of-emotions, body shapes, and body actions). To increase the stability of the model estimate, we standardized each variable. We performed ordinary least squares regression using standardized feature distances to predict trait distances, fitting one model per feature type (i.e., face and body, respectively). We used the adjusted R^2 as the measure of prediction accuracy.

To handle multicollinearity issues across different visual features and increase out-of-sample prediction reliability, we also provided code for performing the regression analysis using Ridge regression with cross-validation (see Supplementary Materials). See codeblock 8.

Step 4: Tests of Significance

We used the Mantel permutation test to evaluate whether our prediction accuracy is better than what we might obtain by chance. For a discussion of the advantages

```

# Test of Research Hypothesis

# Import Package
import pandas as pd
import numpy as np
from sklearn import preprocessing
import statsmodels.api as sm

# Load dependent variable - trait data
# (i.e., trait ratings of images on 1-7 Likert scales)
trait_data = pd.read_csv("sample_trait.csv")
trait_pd = trait_data.iloc[:,1:]
y = trait_pd.values

# Compute representational dissimilarity matrices for trait rating data
# using Euclidean distance
y_RDM = eucRDM(y)

# Define independent variables - different cue types categorized
# as face cues and body cues
face_cues = ['emotion', 'facialStructure']
body_cues = ['shape', 'action']

# Lists to store representational dissimilarity matrices
# for different cue categories
face_x_RDMS = []
body_x_RDMS = []

# Load and process face-related cues
for cue in face_cues:
    face_data = pd.read_csv(f'sample_{cue}.csv')
    face_pd = face_data.iloc[:,1:]
    face_x = face_pd.values
    # Compute representational dissimilarity matrices using Euclidean distance
    face_x_RDM = eucRDM(face_x)
    # Standardize the representational dissimilarity matrices values
    face_x_RDM_st = np.array(preprocessing.scale(face_x_RDM))
    face_x_RDMS.append(face_x_RDM_st)
face_x_RDMS = np.array(face_x_RDMS).T

# Load and process body-related cues
for cue in body_cues:
    body_data = pd.read_csv(f'sample_{cue}.csv')
    body_pd = body_data.iloc[:,1:]
    body_x = body_pd.values
    # Compute representational dissimilarity matrices
    # using appropriate distance measure
    if cue == 'action':
        body_x_RDM = cosRDM(body_x)
    else:
        body_x_RDM = eucRDM(body_x)
    # Standardize the representational dissimilarity matrices values
    body_x_RDM_st = np.array(preprocessing.scale(body_x_RDM))
    body_x_RDMS.append(body_x_RDM_st)
body_x_RDMS = np.array(body_x_RDMS).T

# Perform Ordinary Least Squares regression
model_face = sm.OLS(y_RDM, sm.add_constant(face_x_RDMS)).fit()
model_body = sm.OLS(y_RDM, sm.add_constant(body_x_RDMS)).fit()

# Extract adjusted R-squared values from the fitted models
face_R2 = model_face.rsquared_adj
body_R2 = model_body.rsquared_adj

```

Codeblock 8

of the Mantel permutation test, see Supplementary Materials. To correct for multiple comparisons across models, we used the maximum statistic permutation test. In each permutation, we calculated the test statistics for all tested models, and we selected the maximum statistic to form the null hypothesis distribution, reducing false positives from multiple comparisons.

Specifically, in each permutation iteration, we computed the distance matrix of trait ratings and shuffled its columns and rows using the Mantel permutation procedure. We then refitted the model to obtain new prediction accuracy, repeating this process 2,000 times and obtaining 2,000 prediction accuracies for both face cue and body cue models.

Then, we identified the maximal adjusted R^2 between face-cue and body-cue models in each permutation iteration to generate a null distribution of the maximal statistic across iterations. Based on these respective null distributions, we calculated the significance of the observed prediction accuracy for each cue, using:

$$p = \frac{\text{count}(\text{Adjusted } \tilde{R}_{\max}^2 > \text{Observed Adjusted } R^2) + 1}{\text{count}(\text{Adjusted } \tilde{R}^2) + 1}$$

$\text{Adjusted } \tilde{R}_{\max}^2$ is the maximal adjusted R^2 and $\text{Adjusted } R^2$ is the observed prediction accuracy for face cue model or body cue model. See codeblock 9.

Overall, we find that both face and body feature representations significantly predicted the trait space (face model: adjusted $R^2 = 0.65$, $p < .001$; body model: adjusted $R^2 = 0.03$, $p = .004$). This indicates that the more similar two people's face (or body) features are, the more similarly they are evaluated on warmth. Thus, visual information from both faces and bodies is important to consider in understanding impression formation.

CONCLUSION

Humans' social cognition is complex, and methods that can accommodate this complexity are necessary for understanding. Here, we provide an accessible introduction to an analytical approach that can aid in this endeavor. RSA is a flexible yet powerful way of comparing the representations of different types of data, yet it is relatively novel to social cognitive researchers and therefore underutilized. We hope a better understanding of this technique can advance a more ecologically valid understanding of social cognition.

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```

# Tests of Significance

# Import Package
import math
import random

# Function for Mantel Permutation test
def FlatToMatrix_MantelPermute_MatrixToFlat(data):

    # Get the number of rows and columns in the matrix
    # Given the pre-infinity removal flattened
    # representational dissimilarity matrices
    # Len(Flat_trait) = (x*x-x)/2
    x = int(0.5 + math.sqrt(0.25+2*len(data)))

    # Generate the shuffled row and column IDs
    order = list(range(x))
    random.shuffle(order)

    # Convert the flat list to matrix
    X1 = np.zeros((x,x))
    X1[np.tril_indices(x,-1)] = data # Get the Lower triangle
    X1 = pd.DataFrame(X1) # Convert numpy array to pandas DataFrame
    X1 = X1 + X1.T # Fill out the upper triangle

    # Shuffle the matrix
    X2 = X1.reindex(order) # Reorder the rows based on the shuffled order above
    X3 = X2[order] # Reorder the columns based on the same shuffled order above

    # Flatten the permuted matrix
    # Get the lower-triangle of the permuted matrix
    X4 = list(X3.to_numpy()[np.tril_indices(x,-1)])

    return(X4)

# Number of permutations to perform
iteration_num = 2000

# DataFrame to store results of null distribution
null_R2 = pd.DataFrame()

for iteration in range(iteration_num):
    # Permute the representational dissimilarity matrices of trait data
    y_RDM_perm = FlatToMatrix_MantelPermute_MatrixToFlat(y_RDM)

    # Fit null models for permuted data
    null_model_face = sm.OLS(y_RDM_perm, sm.add_constant(face_x_RDMS)).fit()
    null_model_body = sm.OLS(y_RDM_perm, sm.add_constant(body_x_RDMS)).fit()

    # Store adjusted R-squared values from null models
    iter_pd = pd.DataFrame([[null_model_face.rsquared_adj,
                             null_model_body.rsquared_adj]],
                           columns=['null_face_R2', 'null_body_R2'])
    null_R2 = pd.concat([null_R2, iter_pd], ignore_index=True)

# Extract the maximum R-squared values from each permutation
null_all = null_R2.max(axis= 1)

# Calculate p-values
p_face = (np.sum(null_all > face_R2) + 1) / (len(null_all) + 1)
p_body = (np.sum(null_all > body_R2) + 1) / (len(null_all) + 1)

```

Codeblock 9

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